STRESS SHIFT

A REAL-TIME EMBEDDED CONTROL SYSTEM UTILIZING INVERSE DYNAMICS FOR ADAPTIVE KNEE TORQUE REDISTRIBUTION

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BACKGROUND

INTRODUCTION

Knee conditions affect nearly one quarter of adults (Li) and severely limit quality of life. There is a growing need for preventative technologies that reduce joint stress before surgical intervention becomes necessary. Conventional knee sleeves provide non-targeted support, limiting their ability to respond to changing gaits or individual recovery needs.

CURRENT SOLUTIONS

- Sleeves custom-designed to fit the user, provide support through tailored structure.
- Adaptive materials with varying stiffness or elasticity to provide passive compression
- Expensive and inaccessible
- Rely on passive support, with minimal improvements to active adaptability
- No empirical consensus that they improve knee function

TERMS

- ***Gait analysis*** is the evaluation of human locomotion patterns performed in a research or clinical setting, whether manual or digitized.
- ***Gait cycle*** refers to the sequence of movements that occur when walking, from one step to the next, divided into the stance phase and the swing phase.
- ***Ground reaction force*** is the force exerted by the ground on a body in contact with it (Ou). When the body is moving, the GRF increases past weight due to acceleration forces. It is used to calculate joint torque
- ***Biomechanical torque*** is the rotational force produced around a joint, calculated as force multiplied by the moment arm.

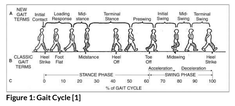


Figure 1: Gait Cycle [1]

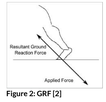


Figure 2: GRF [2]

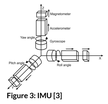


Figure 3: IMU [3]

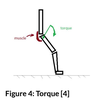


Figure 4: Torque [4]

OBJECTIVES

ENGINEERING PROBLEM

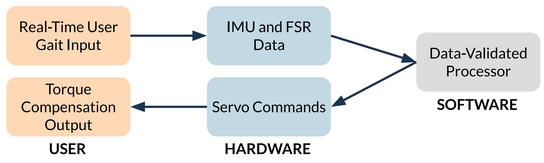
Conventional knee sleeves offer uniform support and cannot provide targeted assistance for complex injuries. This limits their ability to respond to changes in gait or individual recovery needs. Current technological solutions are costly, inaccessible, and unfit for everyday use.

RESEARCH QUESTION

Can a dynamically adaptive knee sleeve be designed to redistribute loads, reducing elevated knee torque at critical gait phases while remaining low-cost, accessible, and suitable for everyday use?

SOLUTION

By collecting gait-specific inertial and pressure data from the leg using IMUs and FSRs, modelling joint torques using GRFs and rotational kinematics, and applying mechanical compensation using lightweight actuators, Stress Shift is designed to selectively intervene when abnormal stress patterns are detected, while remaining flexible during normal motion. Instead of relying on passive load redistribution, Stress Shift operates as an actively adaptive system.



ENGINEERING GOALS

Reduce peak sagittal-plane knee torque by **20%** compared to standard braces.

- Cost: ≤ **60%** of the price of common "smart" knee systems.
- Accessibility: Use only off-the-shelf parts with ≤**2** suppliers per key component. No custom machining.
- Everyday Use:

Weight under **250g**	≤**6** minute setup time	≤**2** minute take-off time	No mobility restrictions

METHODOLOGY

MATERIALS

- MPU 6886 6-Axis IMU (2)
- Force-sensing resistor (2)
- 25kg High-torque RC Servo (4)
- Half Breadboard (1)
- ESP32 Dev Module (2)
- 5v Battery Pack/Power Bank (1)
- Neoprene Knee Sleeve (1)
- Nylon Webbing Strap (4)
- USB Cord (2)
- Small Container with Slots (1)
- Wiring, Velcro, and Adhesive
- Apps: Python, Arduino IDE, Excel

SYSTEM DESIGN



Photo 1: Calf box, Photo taken by presenter using iPhone, 2026

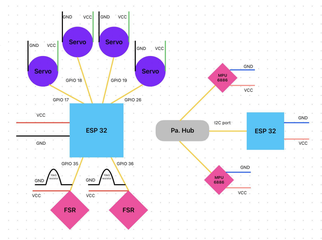


Figure 5: Circuit Schematic, Figure created by presenter using Freeform, 2026

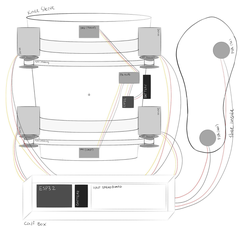


Figure 6: Full Diagram, Figure created by presenter using Procreate, 2026

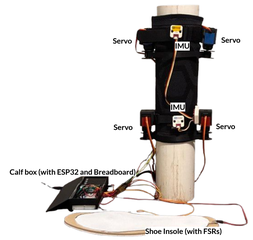


Photo 2: Full Sleeve on Display, Photo taken by presenter using iPhone, 2026

- 2 ESP32s with Wi-Fi UDP
- On ESP32 #1, 2 FSRs (heel, toe)→ADC
- On ESP32 #2, 2 IMUs (thigh, shank)→I2C
- Heel and toe sensor positions defined in set coordinates
- IMUs aligned to sagittal plane

- Two servos on the thigh, two servos on the shank; Controlled through signals from ESP32
- Spool attachments on the servos control nylon straps that tighten around the leg

PROCESSING PROGRAM

IMU KINEMATICS

- Segment orientations for thigh and shank calculated using a complementary filter that combined gyroscope and accelerometer data:
  $\theta(t) = \alpha\left[\theta(t-\Delta t) + \omega\Delta t\right] + (1-\alpha)\theta_{\text{acc}} \quad \alpha = 0.98$
- Knee angle: $\theta_{\text{knee}} = \theta_{\text{shank}} - \theta_{\text{thigh}}$, Angular velocity: $\frac{d\theta_{\text{knee}}}{dt} = \omega_{\text{shank}} - \omega_{\text{thigh}}$
- Low-pass filtering and numerical differentiation are used to calculate the angular velocity and acceleration of the knee joint

GRF AND STANCE DETECTION

- Heel and toe pressure sensor data were converted into vertical ground reaction force (GRF). After scaling to body weight, force scaling during stance phase was determined using: $F_z = k_{\text{tot}}\left(\frac{mg}{\text{mean(rel}_{\text{stance}})}\right) + t_{\text{ot}}\left(\frac{mg}{\text{mean(rel}_{\text{stance}})}\right)$
- Center of pressure: $x_{\text{cop}} = \frac{F_h x_h + F_t x_t}{F_h + F_t}$
- Stance phase detection was defined by vertical force thresholds:
- Stance ON when: $F_z > 0.15\,mg$, Stance OFF when: $F_z < 0.08\,mg$

TORQUE PROXY COMPUTATION

- A simplified sagittal-plane biomechanical model was used to calculate knee joint torque. Anthropometric relationships for the shank were defined as:
  $m_2 = 0.0464m, \quad a_2 = 0.43l_2, \quad I_2 = m_2(0.30l_2)^2$
- The torque proxy was then calculated:
  $\tau_{\text{knee}} = I_2\frac{d^2\theta_{\text{knee}}}{dt^2} + m_2 g a_2 \sin(\theta_{\text{knee}}) - F_z x_{\text{cop}}$
- Torque values were smoothed using a moving average filter (τ_f)

PERSONALIZED THRESHOLDS AND CONTROL LOGIC

- Baseline torque levels for each participant were learned during normal walking. Separate medians were computed for extension and flexion torque: $\text{base}_{\text{pos}} = \text{median}(\tau_f[\tau_f > 0]), \text{base}_{\text{neg}} = \text{median}(\{|\tau_f[\tau_f < 0]\})$
- Engagement thresholds were defined as: *ON* $\rightarrow 1.6 \times \text{base}$, *OFF* $\rightarrow 1.3 \times \text{base}$
- Assist levels were scaled proportionally to torque magnitude:
  $\text{level} = \frac{\tau_f - 1.6\cdot\text{base}_{\text{pos}}}{1.6\cdot\text{base}_{\text{pos}}}, \text{level} = \frac{|\tau_f| - 1.6\cdot\text{base}_{\text{neg}}}{1.6\cdot\text{base}_{\text{neg}}}$

COMMAND EXECUTION

If torque exceeded safe thresholds, the processor generated commands for mechanical servo assistance. High extension torque triggered flexion assistance, while high flexion torque triggered extension assistance. Commands were transmitted in real time to the ESP32 via UDP as NEUTRAL, FLEX, or EXT, each scaled from 0 to 1. Rate limiting and watchdog timeout enforce safety.

DATA

PARTICIPANTS

10 able-bodied, anonymous participants, ages 18-65, mean height 1.77m, mean weight 68kg. Same participants for data collection and testing.

DATA COLLECTION PROCESS

Data collection wearing prototype with servos OFF. 5 walking trials per participant, 50 seconds per trial. Total = 42 minutes of gait data. Sampling frequency ~100Hz.
Data streamed from both ESP32s via Wi-Fi UDP, stored in CSV files.
Raw Data: Timestamp, linear acceleration, angular velocity, FSR pressure.
Compute pressure baselines: $h_{\text{rel}} = \max(0, p_h - h_0)$, $t_{\text{rel}} = \max(0, p_t - t_0)$
Median filtering and data normalization used to reduce noise.

TESTING PROCESS

Same protocol as the data collection process. Standard Trials: Walking trials without servos engaged. Adaptive Trials: Identical walking protocol with servos engaged.
Evaluation Metrics (Returned from processing program): Ground force, torque, filtered torque, stance phase. Standard vs adaptive evaluated through percent reduction in peak torque: $\frac{\tau_{\text{peak, standard}} - \tau_{\text{peak, adaptive}}}{\tau_{\text{peak, standard}}} \times 100$

CONCLUSIONS

CONTRIBUTIONS

Stress Shift reduced peak sagittal-plane knee torque by **32.09%** compared to standard braces, exceeding the 20% target.

- Cost: Stress Shift comes in at $150, **37.5%** of the cost of similar adaptive sleeves on the market starting at $400.
- Accessibility: All parts were easily accessible for purchase, with **1** supplier per key component and only off-the-shelf materials.
- Everyday Use: The finished product took approx. **2.5** minutes to put on and **1.5** minutes to remove, with a weight of **246g**.

Stress Shift demonstrates a low-cost, actively adaptive system for sagittal-plane knee torque redistribution, a novel usage of mechanical compensation with applications in orthopedics, physical therapy, and at-home recovery.

LIMITATIONS AND FUTURE WORK

Limitations: Lack of laboratory access, absence of professional mentorship, and limited funding for testing and prototyping.

- Passive Torque Compensation: The current model does not address passive torques that occur within the joint, only torques that are applied to the joint from 'external' forces. Passive torques can be estimated using IMUs positioned closer to the knee, along with more complex computational physics, for a more accurate, effective device.
- Diversified Dataset: My project relies on data from 10 anonymous participants. For a more scalable product, the dataset can be expanded to include more participants with varied physical demographics.
- Multi-Actuator Integration: The current model uses four servos to provide active torque redistribution around the joint. Future models can experiment with more subtle, targeted actuators such as shape memory alloy strips and mini linear, motor-based actuators.

RESULTS

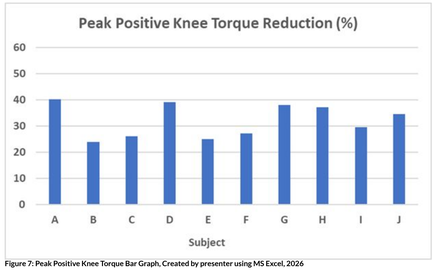


Figure 7: Peak Positive Knee Torque Bar Graph, Created by presenter using MS Excel, 2026

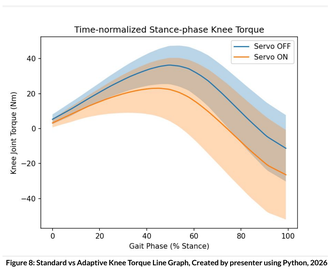


Figure 8: Standard vs Adaptive Knee Torque Line Graph, Created by presenter using Python, 2026

Subject	FLEX	EXT
A	114	38
B	96	42
C	112	40
D	102	40
E	106	50
F	96	42
G	92	50
H	104	30
I	110	34
J	104	44
K	94	50

Figure 9: FLEX and EXT Table, Created by presenter using MS Excel, 2026

Detection to assist activation occurred within one control cycle (≤**10 ms** resolution), indicating near-immediate algorithmic response to torque threshold crossing.

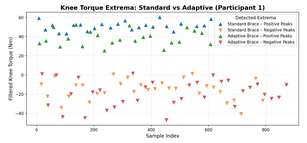


Figure 10: Participant 1 Knee Torque Extrema, Created by presenter using Python, 2026

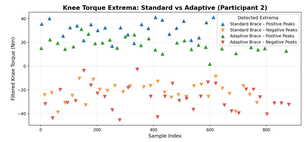


Figure 11: Participant 2 Knee Torque Extrema, Created by presenter using Python, 2026

ANALYSIS

The adaptive sleeve produced a mean **32.09% ± 6.36%** reduction in peak positive knee torque across all **10** participants during the stance phase, with zero adverse events. Despite an anthropometric range spanning **57-92 kg** body mass and **159-182 cm** height, peak reductions remained within **23.4%-40.2%**, indicating the system generalizes accurately across body types. Flexion and extension trigger counts were similar across participants, indicating consistent control behavior.

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RESEARCH

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VISUAL

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